

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Task - 1

Roll No.: T016	Name: Himani Malankar
Class: TYIT	Batch: 1
Date of Assignment: 22/07/2026	Date/Time of Submission: 22/07/2026

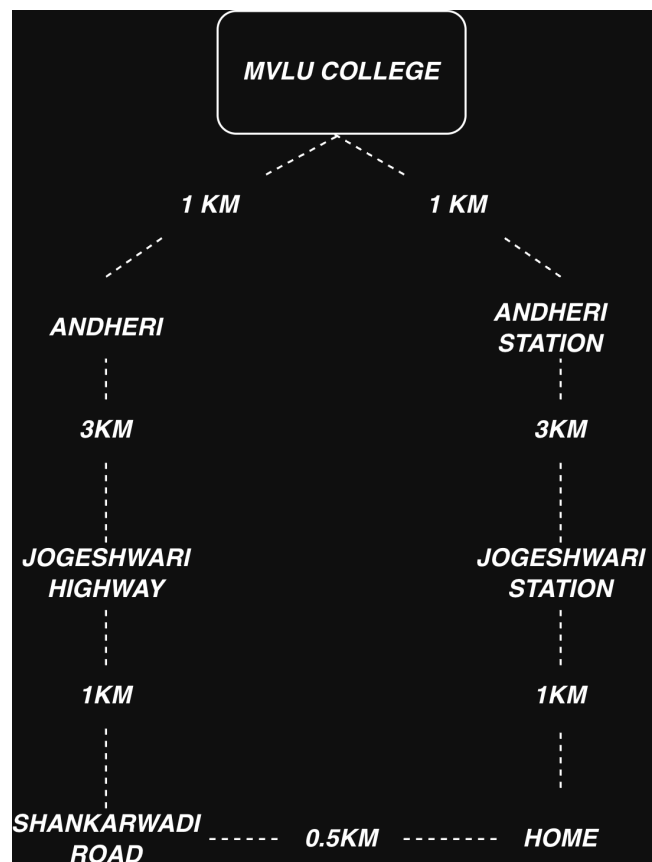
PROBLEM STATEMENT:

Write Python programs to determine the route from your College to your Home using the following search algorithms:

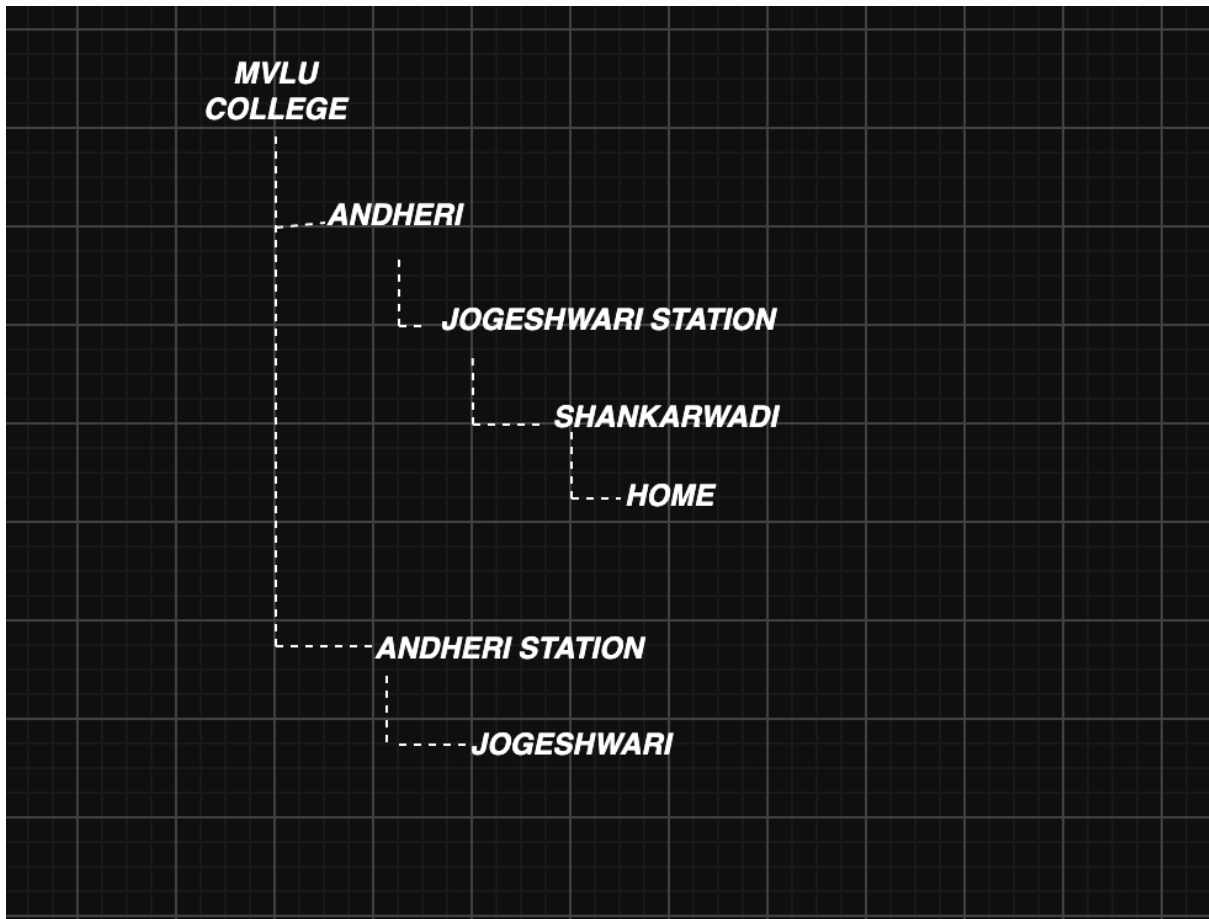
- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Uniform Cost Search (UCS)
- Greedy Best-First Search
- A* Search

Include the graph used in your implementation along with edge costs and heuristic values if required.

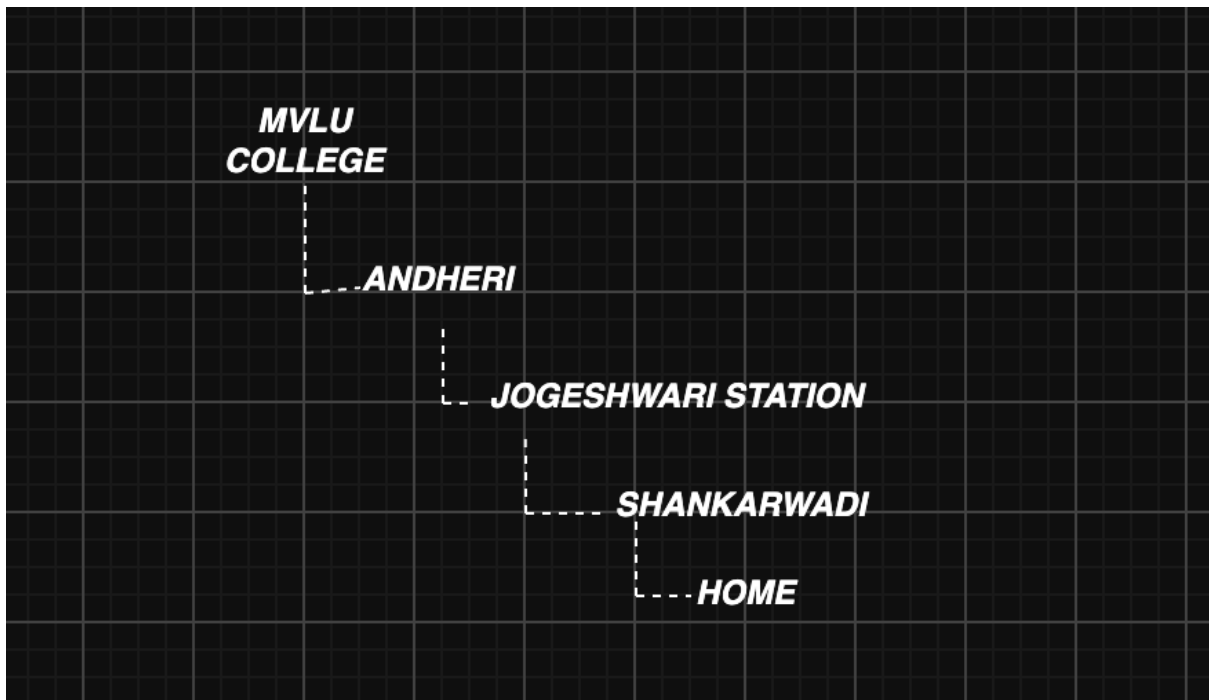
GRAPH:



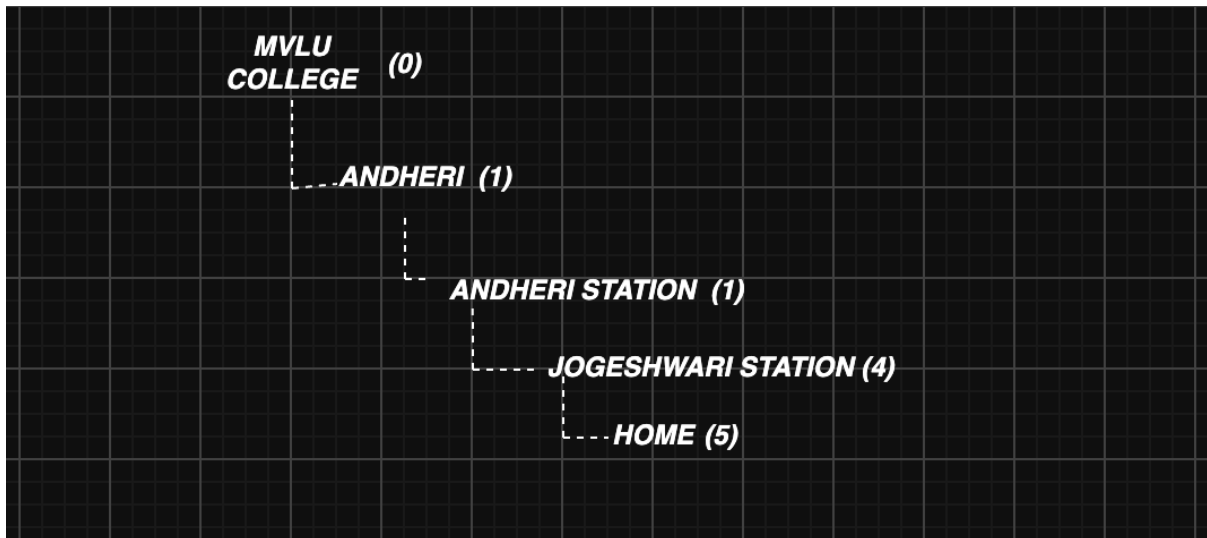
BSF GRAPH:



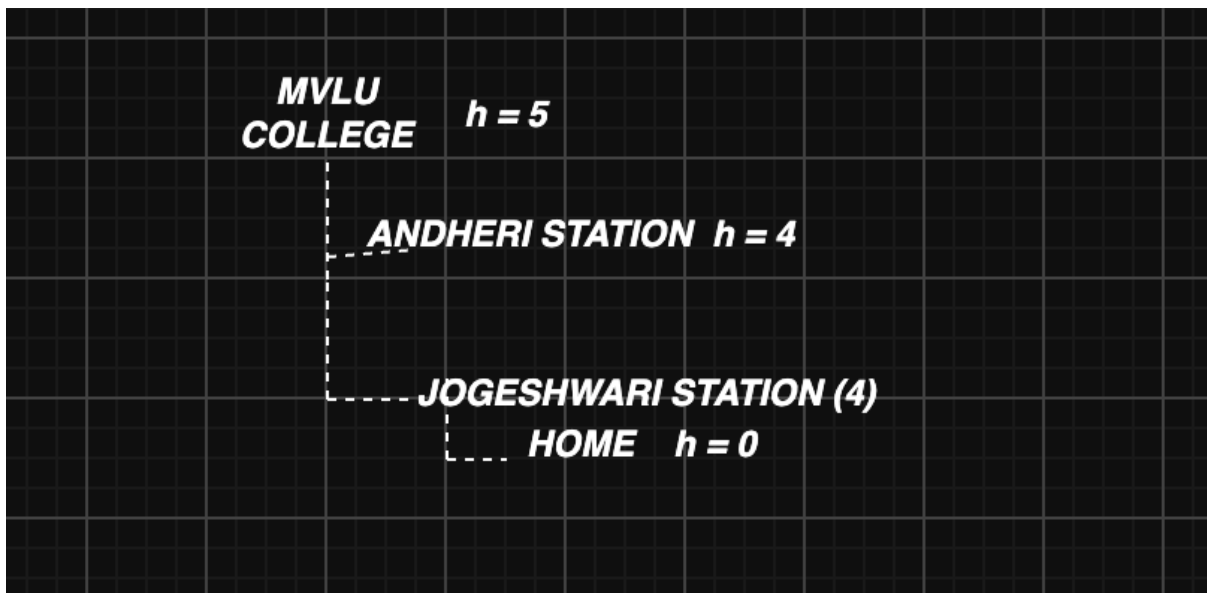
DSF GRAPH:



UCS GRAPH:



GREEDY BEST-FIRST SEARCH GRAPH:



A * SEARCH ALGORITHM: